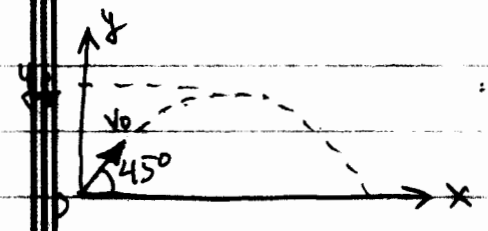
Solutions to HW07 (1) Ch.3 S&F Starykh

3.25



height $y_f = 12\,ft = 3.66\,m$; $y_i = 0$

$1\,ft = 0.3048\,m$

At this highest point $v_y = 0$

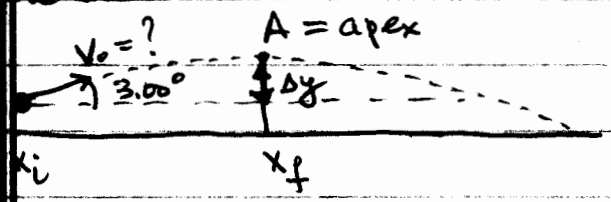
$\Rightarrow$ use $v_y^2 = v_{y0}^2 - 2g(y_f - y_i) = 0$

$$\Rightarrow v_{y0}^2 = 2g(y_f - y_i) \Rightarrow v_{y0} = \sqrt{2g(y_f - y_i)}$$

$$v_{y0} = v_0 \sin 45°$$

$$\Rightarrow v_0 = \frac{\sqrt{2g(y_f - y_i)}}{\sin 45°} = 11.97\,m/s = \boxed{12\,m/s}$$

3.27 Side view



$\Delta y = y_A - y_i = 0.330\,m$

$\Delta x = x_f - x_i = 12.6\,m$

$$v_{y0} = v_0 \sin(3.00°)$$

$$v_{x0} = v_0 \cos(3.00°)$$

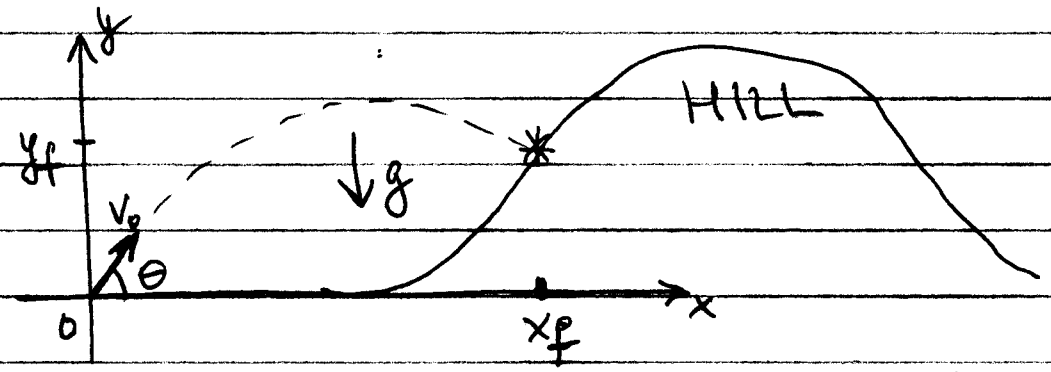
At the apex A $v_y^2 = 0 = v_{y0}^2 - 2g\Delta y$

$$\Rightarrow v_{y0} = \sqrt{2g\Delta y} \Rightarrow v_0 = \frac{\sqrt{2g\Delta y}}{\sin(3.00°)} = 48.59\,m/s$$

$$\boxed{v_0 = 48.6\,m/s}$$

Note that information on Δx displacement was not used.

3.28



$v_0 = 300\,m/s$, $\theta = 55.0°$, time of flight $t = 42.0\,s$

Choose $x_i = y_i = 0$, then $\Delta x = x_f$, $\Delta y = y_f$

Along x: motion with constant velocity $v_{x0} = v_0 \cos\theta$

$$\Rightarrow \Delta x = v_0 \cos\theta \cdot t = 300\,m/s \cdot \cos 55.0° \cdot 42.0\,s = 7227\,m$$

Along y: motion with constant acceleration $a_y = -g$

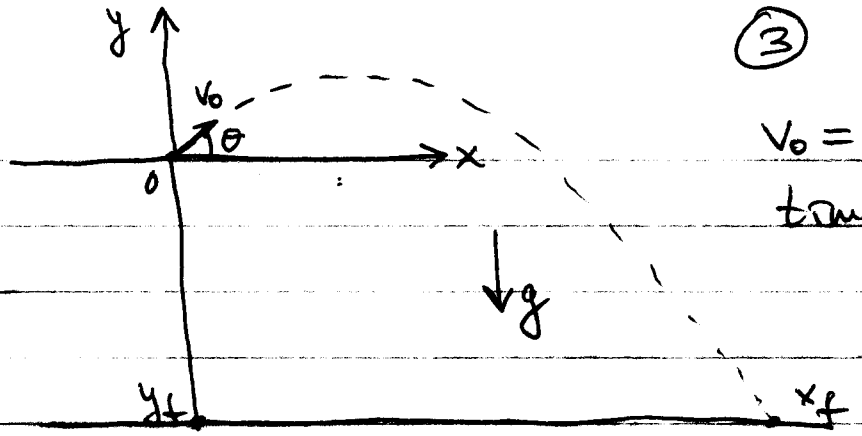
$$\Rightarrow \Delta y = v_{y0} t - \frac{1}{2} g t^2 = (300 \cdot \sin 55.0°)(42.0) -$$

$$- \frac{1}{2}(9.80)(42.0)^2 = 10321 - 8643 = 1677\,m$$

The answer to 3 significant figures

$$\boxed{\begin{aligned} x_f &= 7230\,m \\ y_f &= 1680\,m \end{aligned}}$$

3.29



$V_0 = 15\ m/s$, $\theta = 25^0$

time of flight

$t = 3.0\ s$

In the chosen system of coordinates $x_i = 0$, $y_i = 0$

$$y_f - y_i = V_{y_0} t - \frac{1}{2} g t^2$$

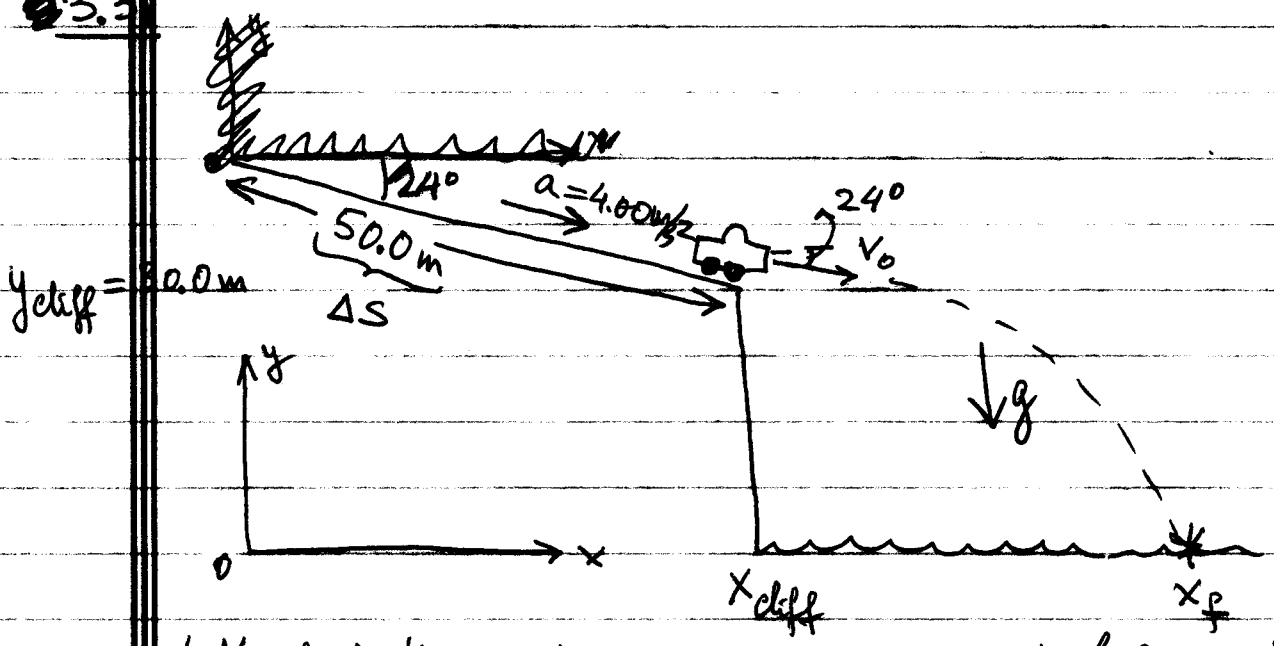
$$V_{y_0} = V_0 \sin\theta = 15\ m/s \cdot \sin 25^0 = 6.34\ m/s$$

$$\Rightarrow y_f - y_i = 6.34 \cdot 3 - \frac{1}{2} 9.8 \cdot (3.0)^2 = 19.02 - 44.1 =$$

$$= -25.08\ m$$

The height of the building $h = |y_f - y_i| = \boxed{25\ m}$

3.3



Let's find the speed of the car, v_0, just before it flies off the cliff : $v_0^2 = v_i^2 + 2a\Delta s$

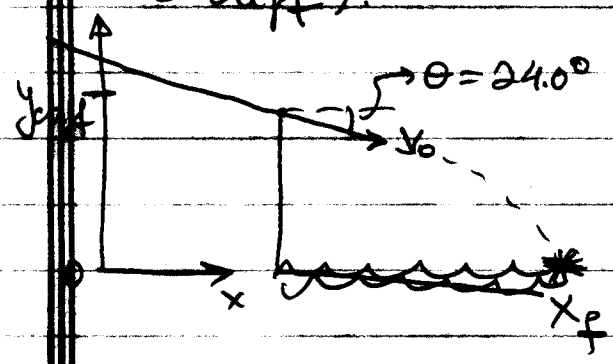
$V_i = 0$ (when car is parked)

$\Delta S = 50.0\,m$ distance from the parking spot to the edge of the cliff

$\Rightarrow v_0^2 = 2 \cdot 4.00\,m/s^2 \cdot 50.0\,m = 400\,m^2/s^2$

$\Rightarrow v_0 = 20.0\,m/s$

(This is the speed with which the car leaves the cliff).



$y_{ocean} = 0$

$y_{cliff} = 30.0\,m$

$\theta = 24.0°$ below horizontal

$\Rightarrow V_{y0} = -V_0 \sin 24° = -8.13\,m/s$

$V_{x0} = V_0 \cos 24° = 18.3\,m/s$

$$\Rightarrow \Delta y = y_{ocean} - y_{cliff} = V_{y0} t - \frac{1}{2} g t^2$$

$$\Rightarrow \quad -30 = -8.13t - 4.9t^2$$

Solve for t:

$$4.9t^2 + 8.13t - 30 = 0$$

$$t = \frac{-8.13 \pm \sqrt{(8.13)^2 - 4(4.9)(-30)}}{2 \cdot 4.9} = \frac{-8.13 \pm 25.6}{9.8}$$

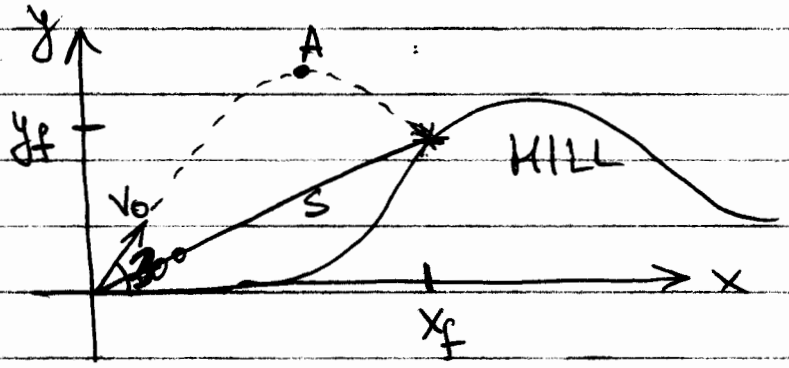
Take <u>positive</u> root: $t = \frac{-8.13 + 25.6}{9.8} = 1.78\,s$

$\Rightarrow$ (b) $\boxed{t = 1.78\,s}$

(a) $x_f - x_{cliff} = V_{x0} t = 18.3 \cdot 1.78 = \boxed{32.6\,m}$

$y_f = y_{ocean} = 0$

3.33



$V_0 = 60.0\ m/s$, $\theta = 30.0°$, time of flight $t = 4.00\ s$

(b) $x_f = V_{x0} t = V_0 \cos\theta\, t = 208\ m$

$y_f = V_{y0} t - \frac{1}{2} g t^2 = V_0 \sin\theta\, t - \frac{1}{2} g t^2 =$

$= 60 \sin 30 \cdot 4 - 4.9(4)^2 = 41.6\ m$

Straight-line distance $s = \sqrt{x_f^2 + y_f^2}$ from the right triangle in the picture.

$s = \sqrt{208^2 + 41.6^2} = \boxed{212\ m}$

(a) Velocity v_A at the highest point A?

Have to decide first if A is reached before the projectile lands on the hill $\Rightarrow$ find time t_A

at A: $v_y = 0 = V_{y0} - g t_A \Rightarrow t_A = \frac{V_{y0}}{g} = \frac{V_0 \sin\theta}{g}$

$t_A = \frac{60 \cdot \sin 30°}{9.8} = 3.06\ s < 4.00\ s$

(time to A < time of flight)

$\Rightarrow$ A is reached before landing

$\Rightarrow V_A = (V_{Ax}, V_{Ay}) = (V_{x0}, 0) = \boxed{(52; 0)\ m/s}$

$V_{x0} = V_0 \cos 30° = 60 \cdot \cos 30° = 52\ m/s$